

Lesson 8: Improper Integrals

Improper Integrals Assessment Rubric

- **Identification (20%):** Correctly identifies the improper integral, recognizing either infinite bounds (e.g., $\int_1^{\infty}$) or vertical asymptotes within the interval.
- **Methodology & Limits (30%):** Correctly sets up the improper integral as a limit, replacing the singularity with a variable (e.g., $t \rightarrow \infty$).
- **Antidifferentiation (20%):** Accurately finds the antiderivative of the function, applying techniques like integration by parts or partial fractions when necessary.
- **Evaluation & Convergence (30%):** Evaluates the limit correctly to determine if the integral converges to a finite number or diverges to infinity.